

POINTERS

Pointer is a variable, which holds the address of another variable of same type.

Pointer is a memory location, which holds the address of another memory location.

Pointer is a derived data type.

Advantages:

1. Dynamic memory allocation.
2. Program performance is increased due to preventing memory wastage.
3. They are very much used in System programming.
4. They are very much used in dynamic linked list & Stacks [data structures].
5. **It allows to access local variable outside the function i.e. data sharing between functions.**
[**call by address/Reference**].
6. **To handle strings, arrays etc in functions we need pointers.**

7. To handle **data files** we are using pointers.
8. They directly works on variable address. Due to this search time is reduced and execution speed is increased.

Dis-advantage:

They are not secured.

Syntax:

datatype * variable;

- * indicates it is a pointer data type.
- * is called indirection operator.
- * is called dereferencing operator.
- *** is a re-direction operator.**
- * indicates value at that address.
- * indicates pointer value.

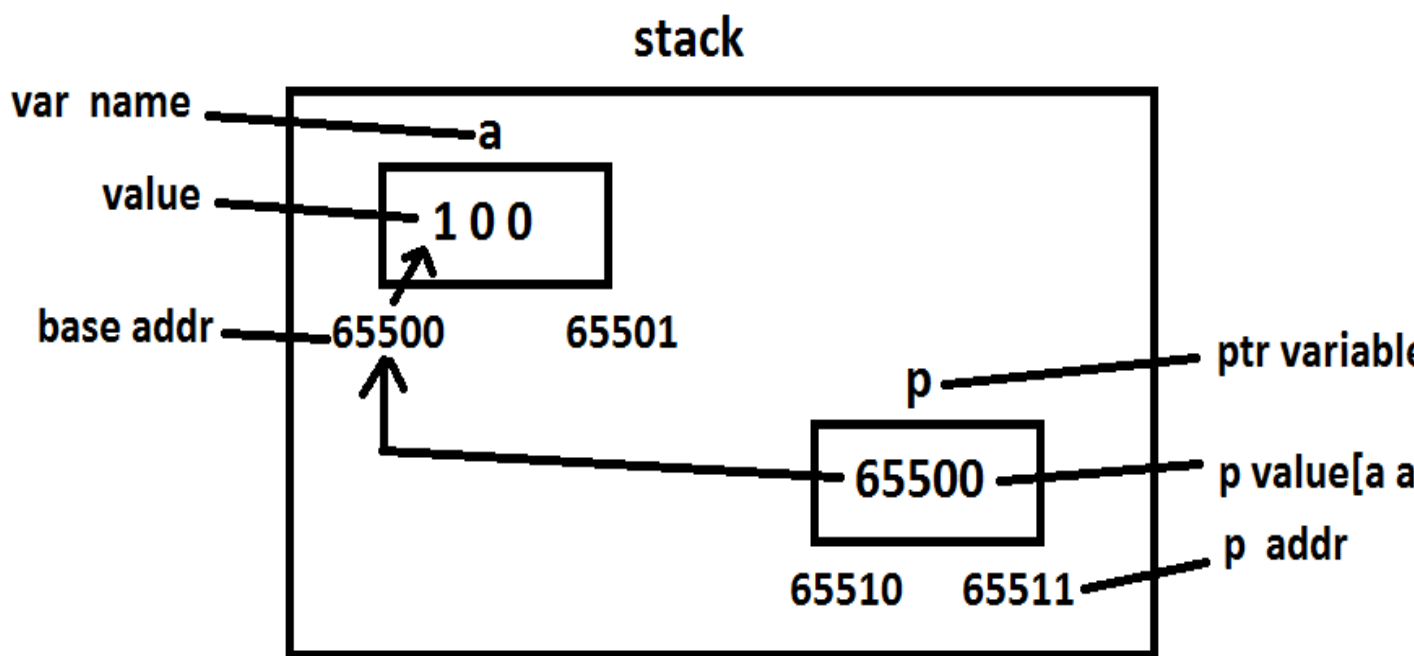
Eg:

```
int a=100, * p;
```

In the above example '**a**' is a general variable.

* indicates '**p**' is a pointer type variable and it is able to store the address of general variable '**a**' as follows.

```
p = &a;
```



In the above example, to pick the value of **a** through pointer variable **p**, we have to use the **printf()** as follows.

```
printf( "%d", *p );
```

→ **100**

Here ***p** means **value of p** or **value at that addr.** i.e. **65500**. But **65500** is the **addr of 'a'**. The **value in a address is 100**.

Or

Here **p** means **65500**. ***p** means **value at 65500**. i.e. **100**.

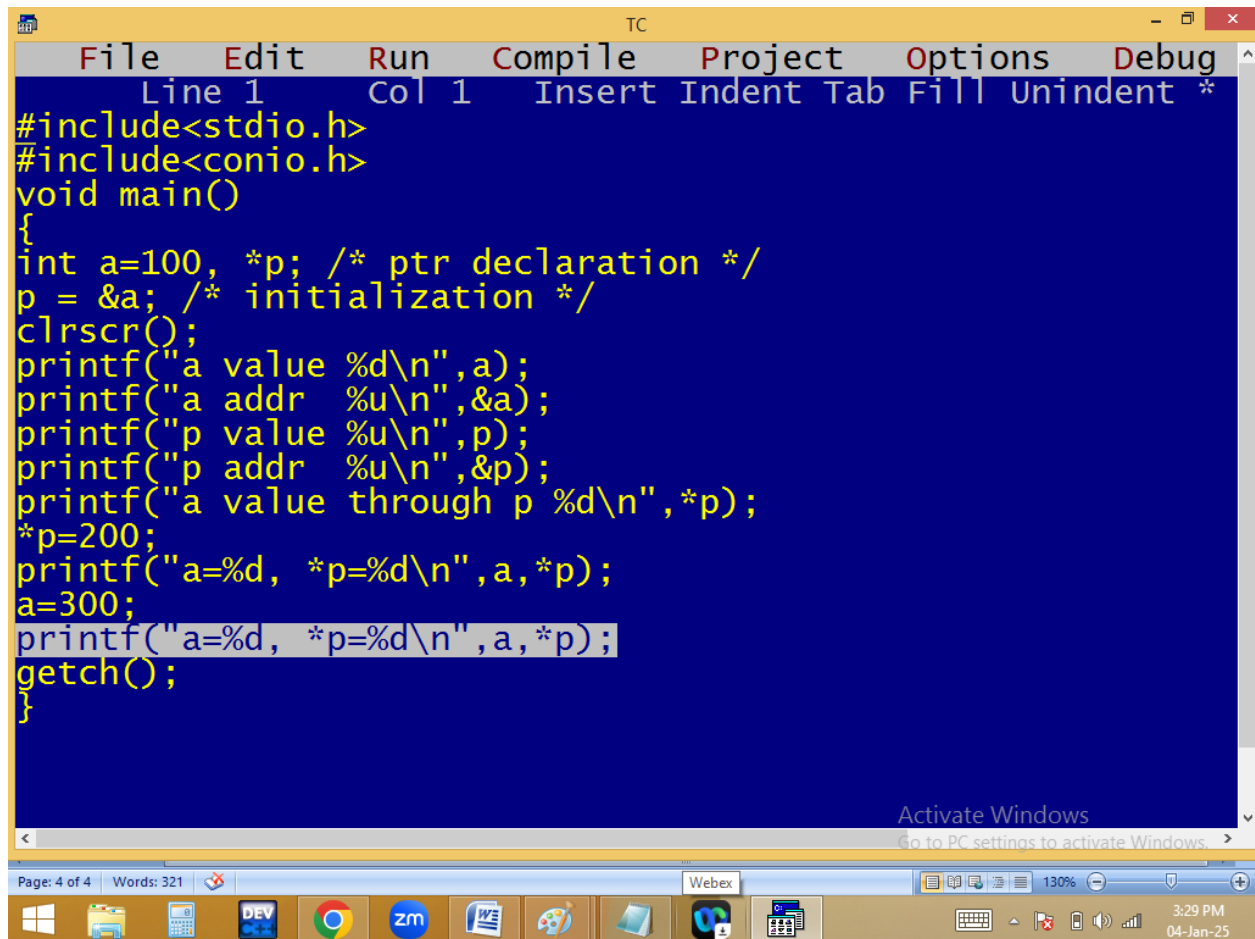
Due to this example any changes conducted in ***p** effects the value of **'a'**. Hence **p** is called **pointer to a**.

Eg: ***p=200;**

Now **a** value becomes **200**.

Eg:

Finding a variable value and address using a pointer:



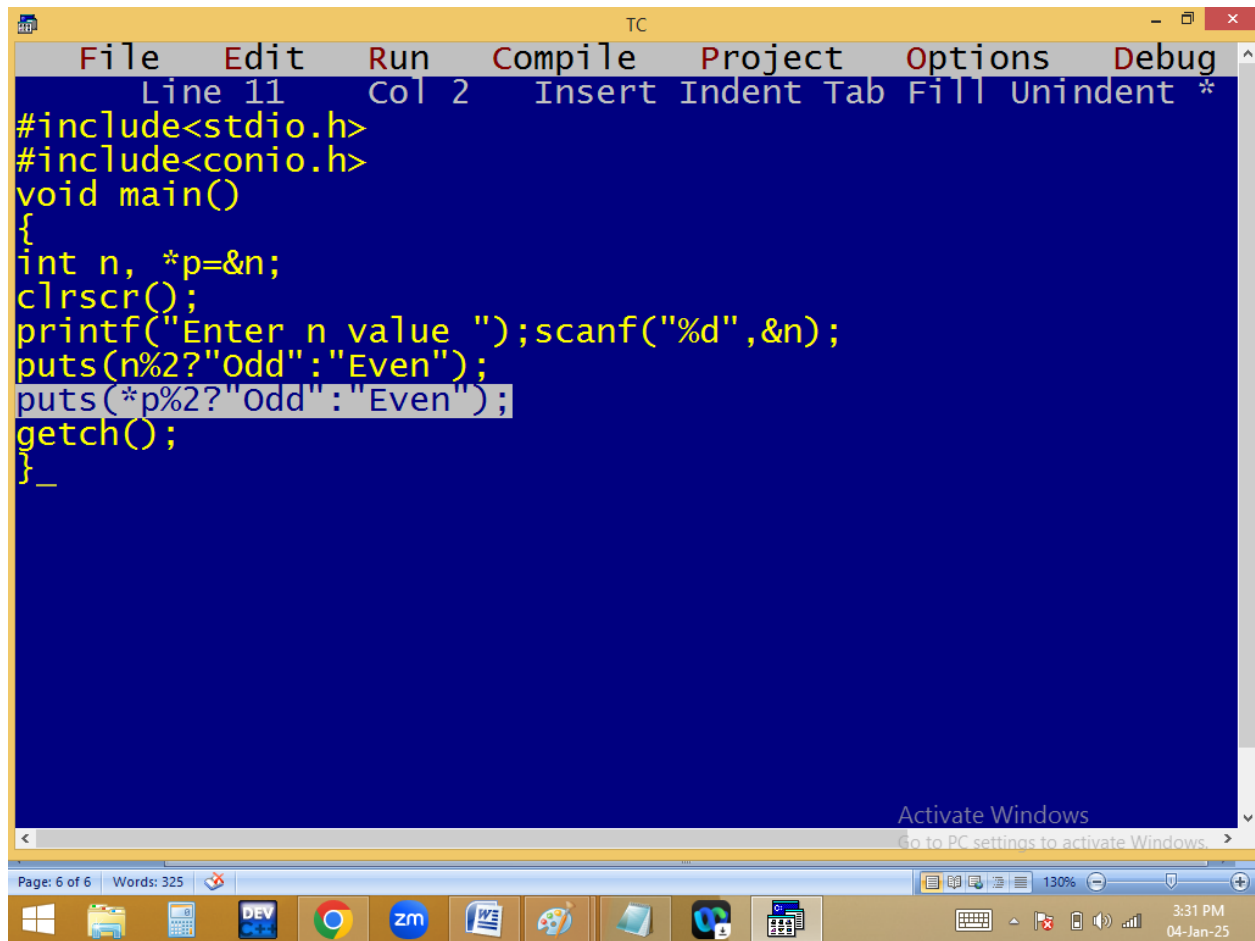
```
TC
File Edit Run Compile Project Options Debug
Line 1 Col 1 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100, *p; /* ptr declaration */
p = &a; /* initialization */
clrscr();
printf("a value %d\n",a);
printf("a addr %u\n",&a);
printf("p value %u\n",p);
printf("p addr %u\n",&p);
printf("a value through p %d\n",*p);
*p=200;
printf("a=%d, *p=%d\n",a,*p);
a=300;
printf("a=%d, *p=%d\n",a,*p);
getch();
}
```

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```
a value 100
a addr 65500
p value 65500
p addr 65502
a value through p 100
a=200, *p=200
a=300, *p=300
```

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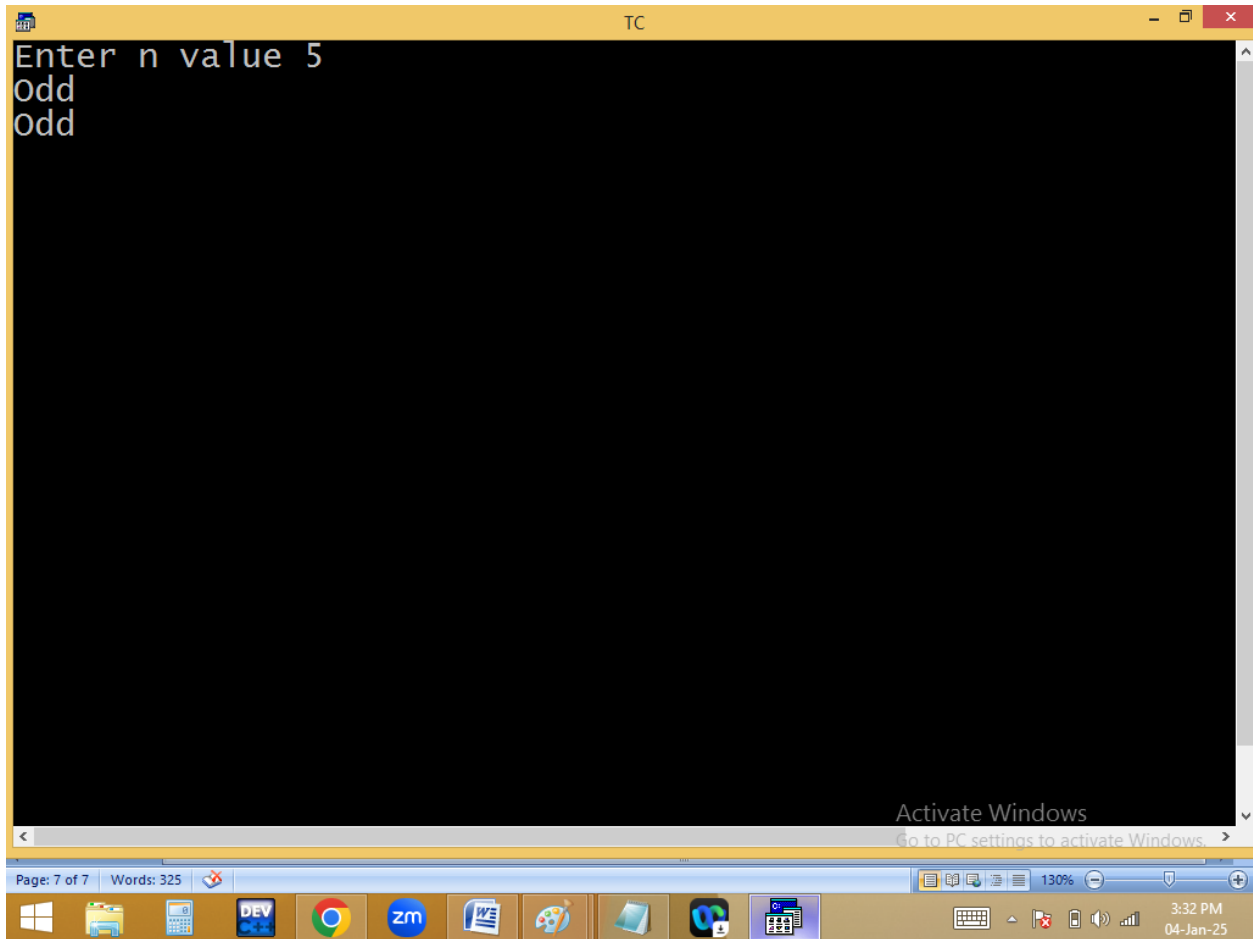
Finding even/odd using pointer:

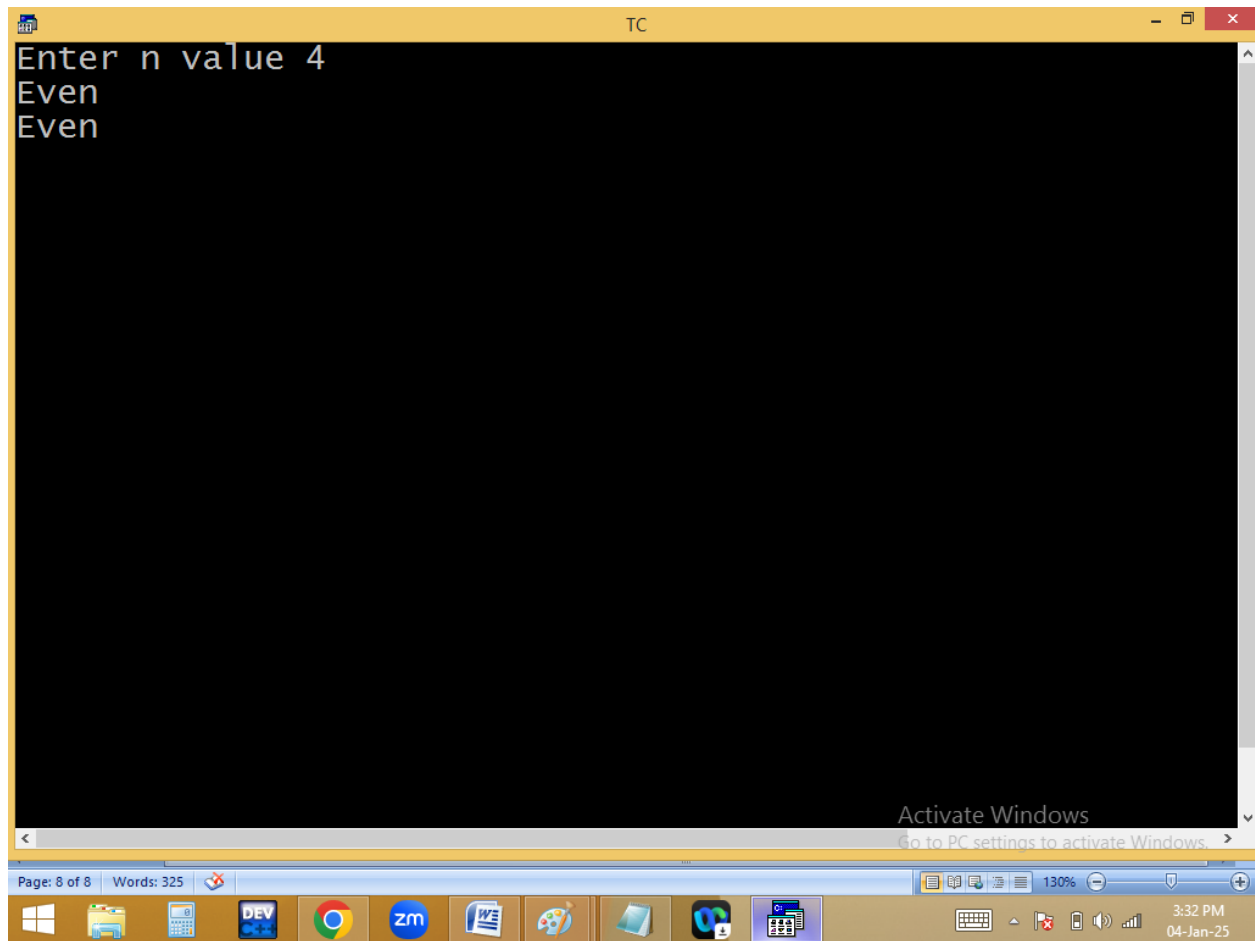


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 11", "Col 2", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main editing area has a dark blue background with yellow text. The code is as follows:

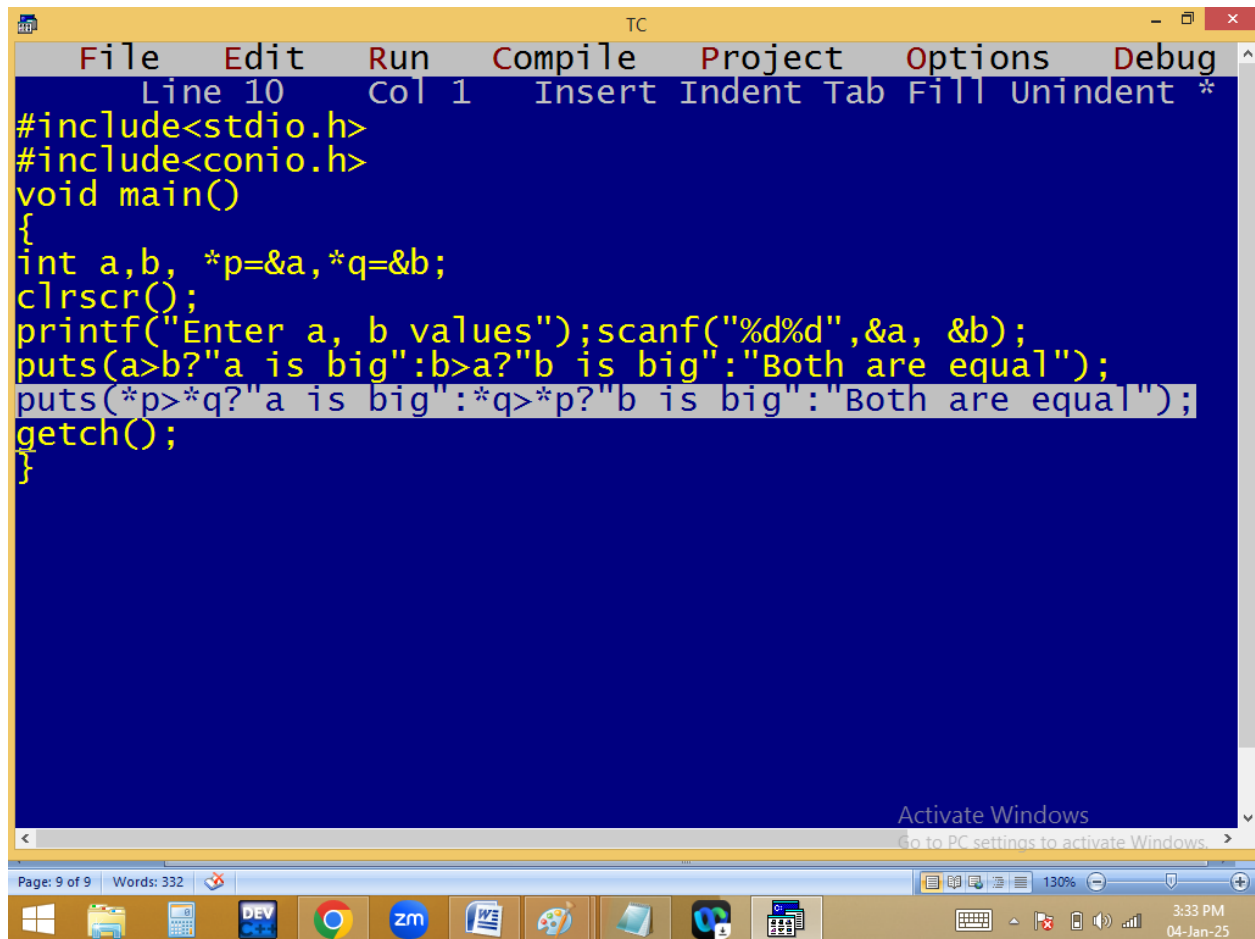
```
#include<stdio.h>
#include<conio.h>
void main()
{
int n, *p=&n;
clrscr();
printf("Enter n value ");scanf("%d",&n);
puts(n%2?"Odd":"Even");
puts(*p%2?"Odd":"Even");
getch();
}_
```

At the bottom of the IDE window, there is a status bar with "Page: 6 of 6" and "Words: 325". Below the IDE window is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and other applications. The system tray on the right shows the time as 3:31 PM and the date as 04-Jan-25. A watermark "Activate Windows" is visible in the bottom right corner of the IDE window.





Finding max in 2 no's using pointers:



```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 1 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b, *p=&a,*q=&b;
clrscr();
printf("Enter a, b values");scanf("%d%d",&a, &b);
puts(a>b?"a is big":b>a?"b is big":"Both are equal");
puts(*p>*q?"a is big":*q>*p?"b is big":"Both are equal");
getch();
}
```

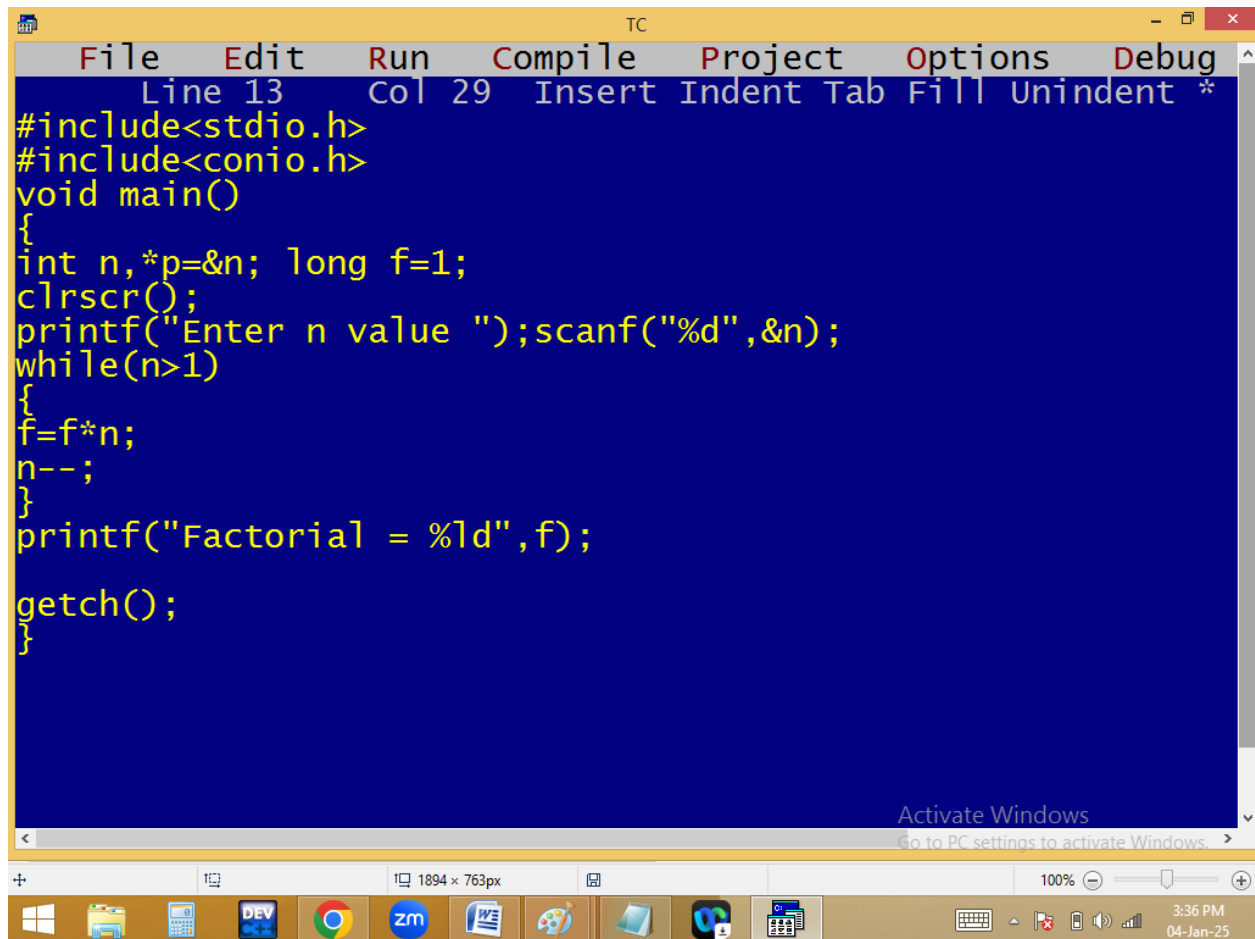
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```
Enter a, b values3 3
Both are equal
Both are equal
```

Activate Windows
Go to PC settings to activate Windows.

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Finding factorial using pointer:

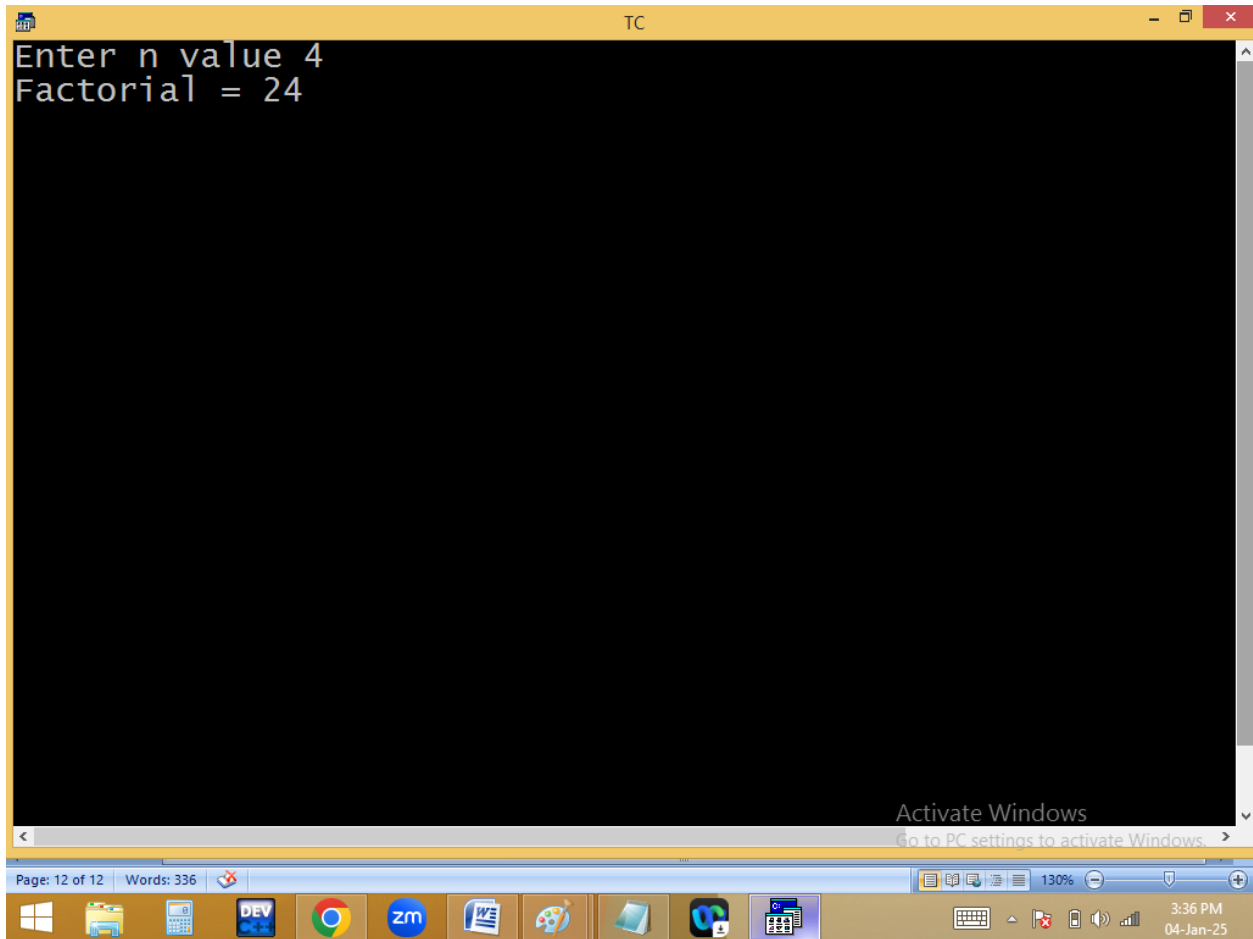


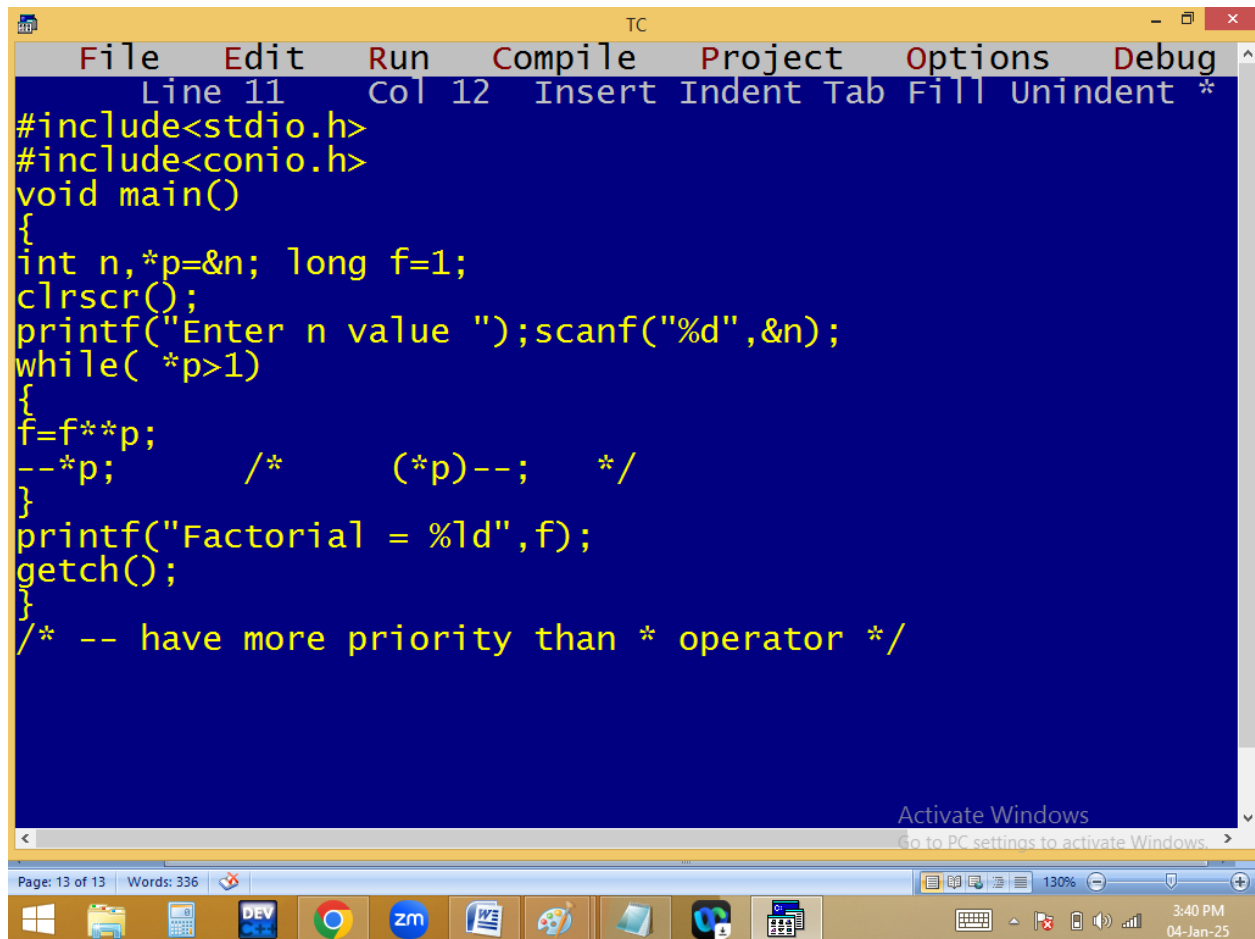
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 13", "Col 29", and "Insert Indent Tab Fill Unindent *". The main editing area has a blue background and contains the following C code for calculating a factorial:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,*p=&n; long f=1;
clrscr();
printf("Enter n value ");scanf("%d",&n);
while(n>1)
{
f=f*n;
n--;
}
printf("Factorial = %ld",f);

getch();
}
```

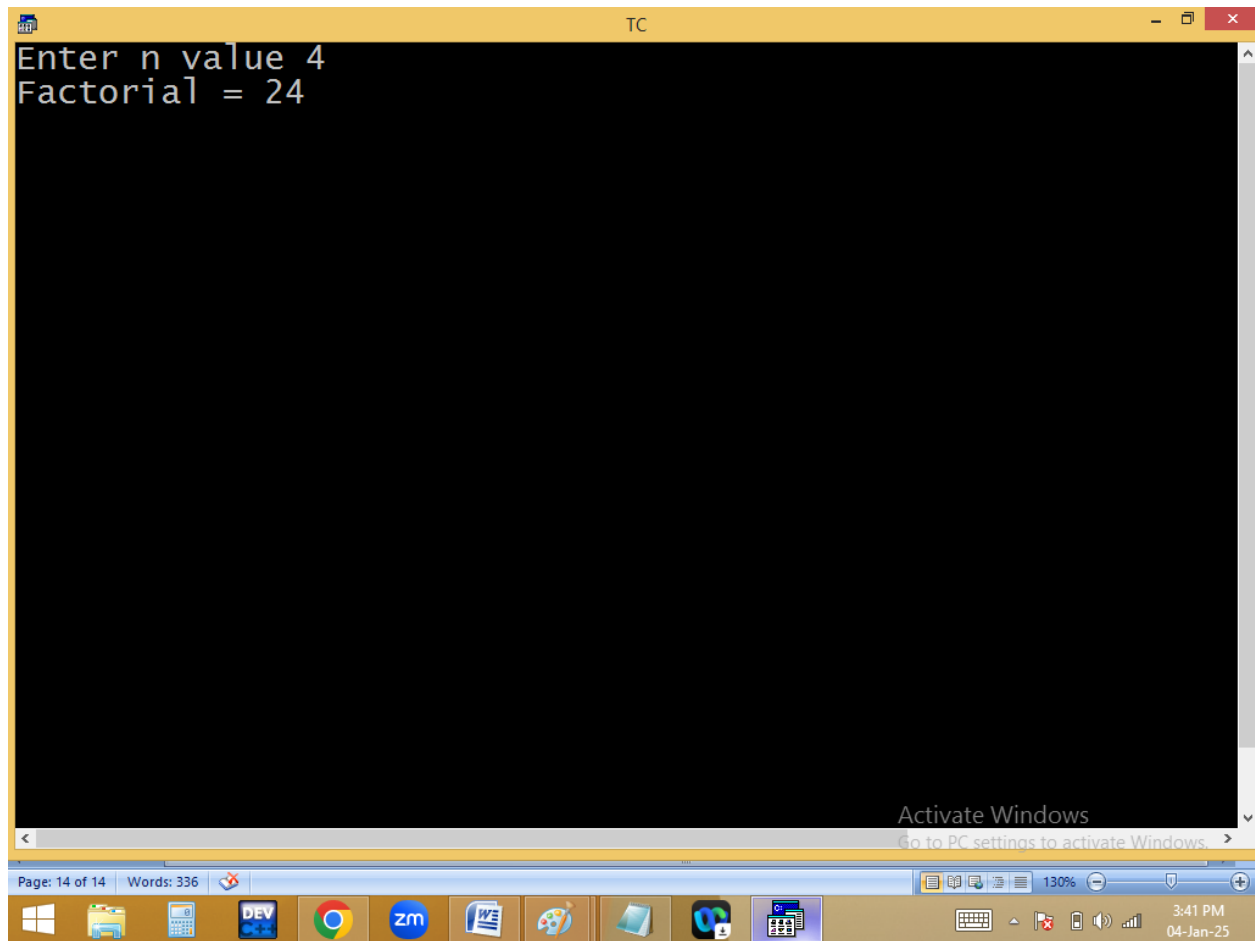
At the bottom right of the editing area, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The taskbar at the bottom shows various application icons including Windows Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and others. The system tray on the right shows the time as 3:36 PM and the date as 04-Jan-25.





```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 12 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int n,*p=&n; long f=1;
clrscr();
printf("Enter n value ");scanf("%d",&n);
while( *p>1)
{
f=f**p;
--*p;      /*      (*p)--;      */
}
printf("Factorial = %ld",f);
getch();
}
/* -- have more priority than * operator */

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130%
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04-Jan-25
```

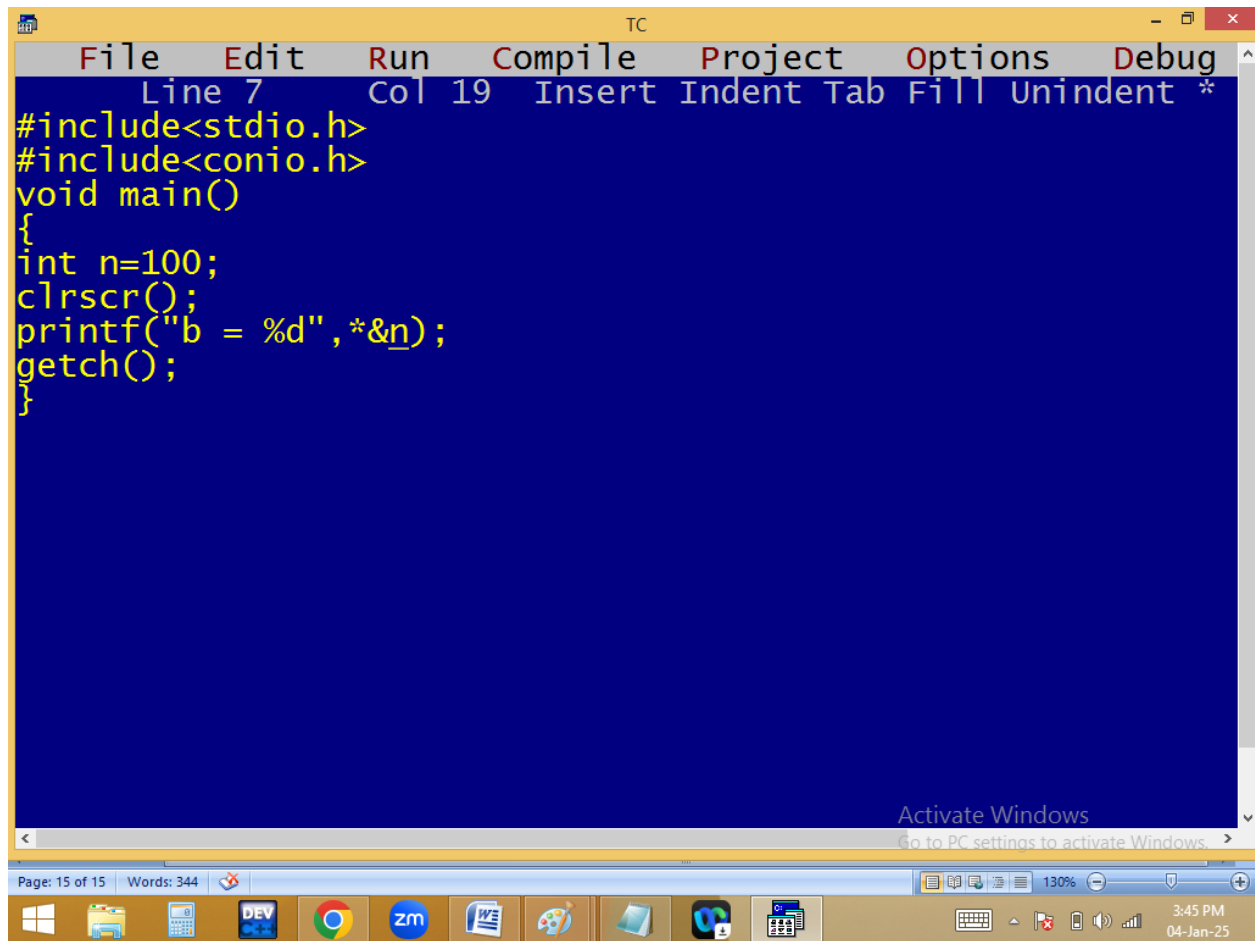


```
Enter n value 4
Factorial = 24
```

Activate Windows
Go to PC settings to activate Windows.

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Finding a normal variable value using pointer mechanism?

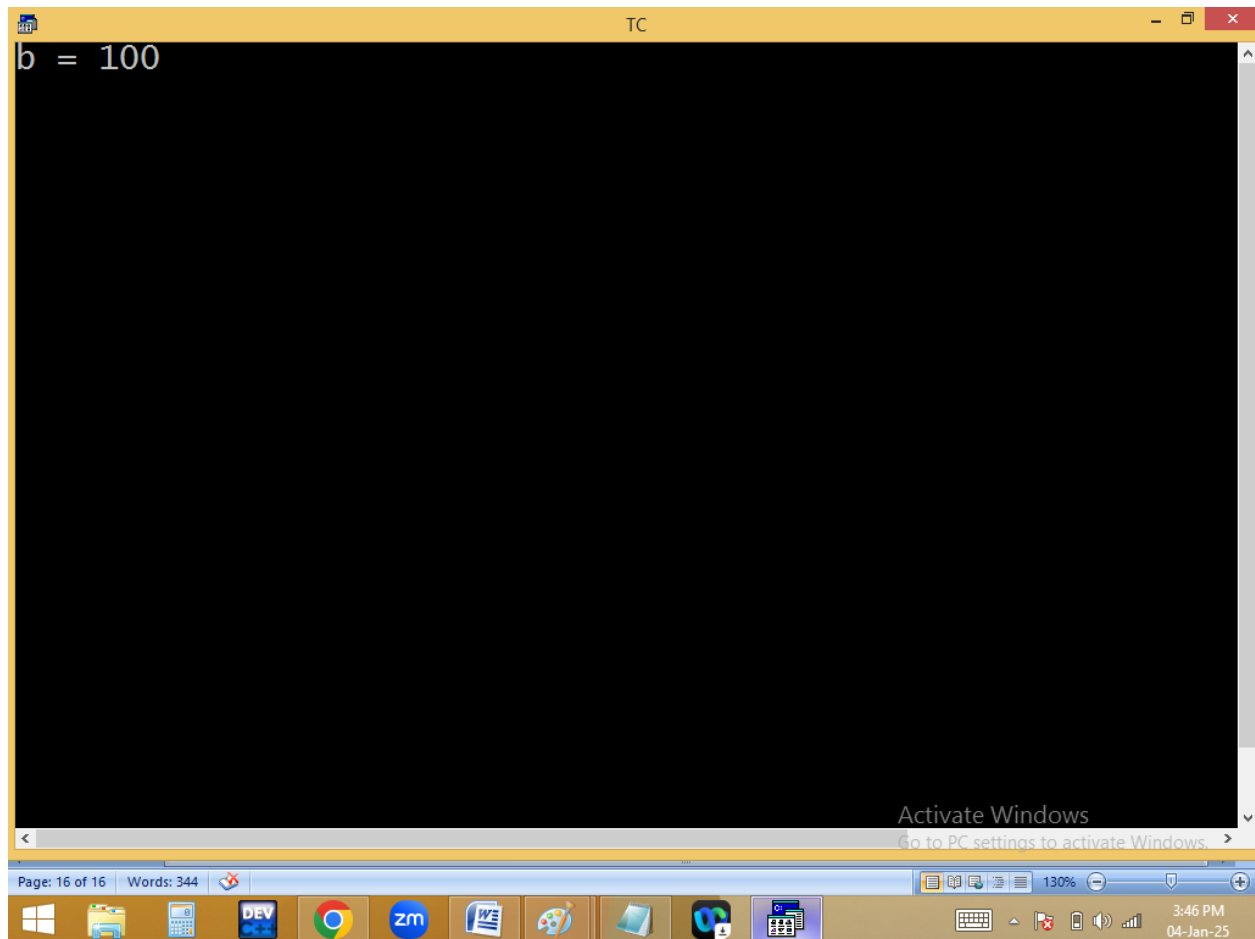


```
TC
File Edit Run Compile Project Options Debug
Line 7 Col 19 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int n=100;
clrscr();
printf("b = %d",*&n);
getch();
}
```

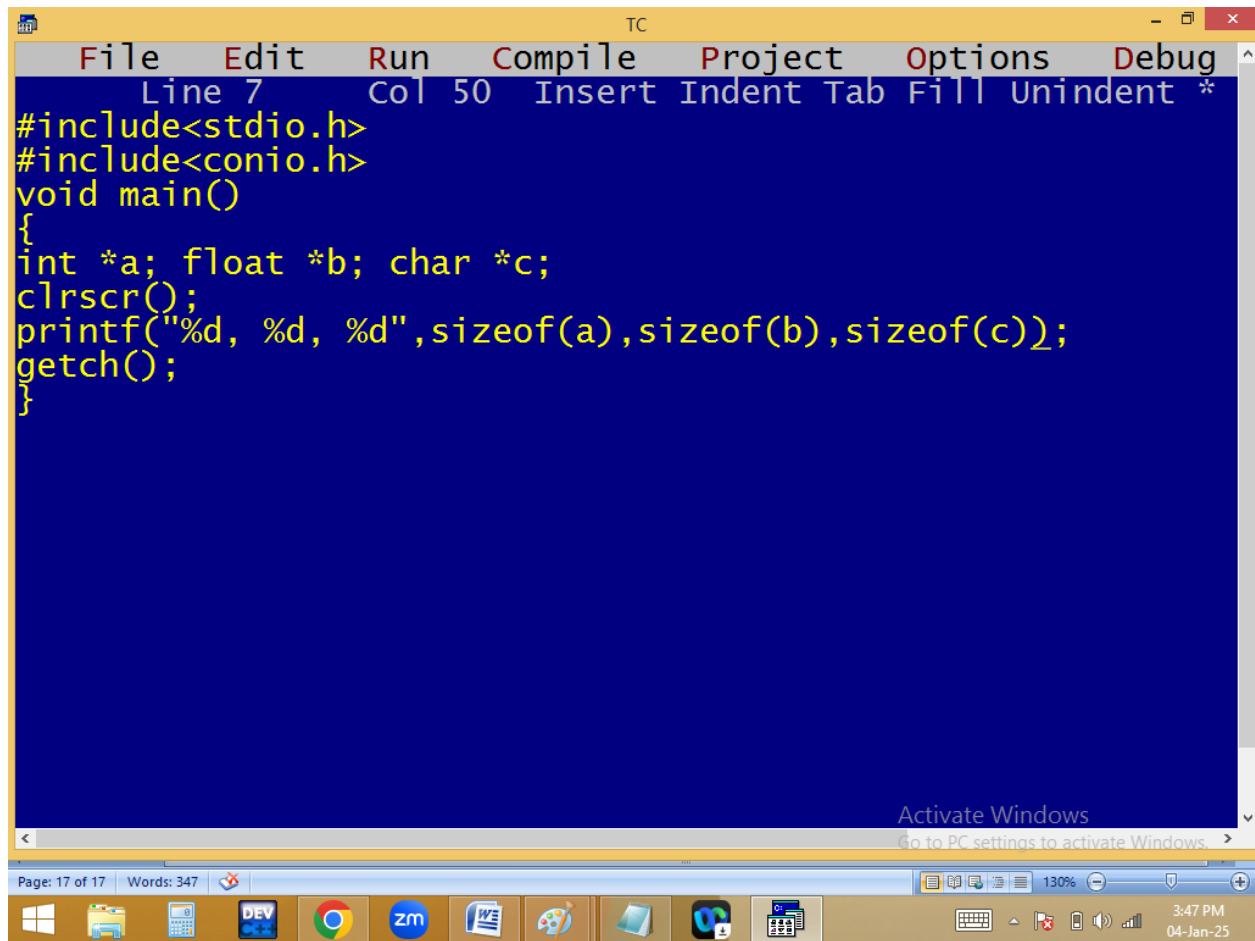
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Activate Windows
Go to PC settings to activate Windows.

3:45 PM
04-Jan-25



Finding pointer size: Any kind of pointer it stores only the address and address always unsigned int. hence pointer takes 2 / 4 / 8 bytes in 16 / 32 / 64 bit compilers.



```
TC
File Edit Run Compile Project Options Debug
Line 7 Col 50 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int *a; float *b; char *c;
clrscr();
printf("%d, %d, %d",sizeof(a),sizeof(b),sizeof(c));
getch();
}
```

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